Exam Date & Time: 06-Dec-2023 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

V SEMESTER B.TECH END SEMESTER EXAMINATIONS, DECEMBER 2023

CIOUD COMPUTING [DSE 3157]

Marks: 50 **Duration: 180 mins.**

A

Answer all the questions.

Instructions to Candidates: Answer ALL questions Missing data may be suitably assumed

1) Analyse the structure of Simple Object Access Protocol (SOAP) in web services. Provide an example to apply its use in real-world applications. (5)

A)

B) With the help of an example, illustrate the significance of a 3-Tier System Architecture in the Service-Oriented Architecture (SOA) framework. (3)

C) Analyse the four principles of REST architecture in the context of distributed systems. (2)

2) Analyse the implementation and strategic implications of Hyper-Converged Infrastructure (HCI) at the organizational level. Discuss how HCI addresses the challenges of traditional infrastructure and evaluate its impact on resource utilization, scalability, and overall data centre efficiency. (5)

A)

B) Justify the requirement of virtualization in cloud computing. Also, illustrate the fundamental characteristics of Virtualization technique. (3)

C) "Live migration needs a shared storage for virtual machines in the server's pool, but cold migration does not." Justify the statement with an example. (2)

3) Summarize the significance of the Enterprise Bus and Publisher-Subscriber Model while establishing the Message-Oriented Middleware (MOM) in distributed systems. Also, justify how MOM reduces the complexity of developing applications that span multiple operating systems and network protocols. (4)

A)

B) Illustrate the working mechanism of the entities involved in the grid of grid services. (3)

C) With the help of an example, justify the need of Gateways in service-oriented architecture. (3)

4)		Analyse the architectural characteristics and advantages of Multiple-Instruction, Multiple-Data (MIMD) systems in the context of parallel computing. Also, differentiate the different types of MIMD systems in terms of architecture, Efficiency, and Scalability.	(5)
	A)		
	B)	Identify the properties of a service-oriented architecture that differentiates it from the distributed systems.	(3)
	C)	Summarize the components of the Workflow system that are involved in the interaction between the services.	(2)
5)		Evaluate the critical role of the CIA (Confidentiality, Integrity, and Availability) in ensuring security within different cloud service models. Also, discuss the possible security attacks associated with each component of the CIA in the context of cloud computing.	(5)
	A)		
	B)	Analyse the different stages of security while building a cloud-specific application.	(3)
	C)	Justify the need of firewalls in any cloud deployment models.	(2)

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